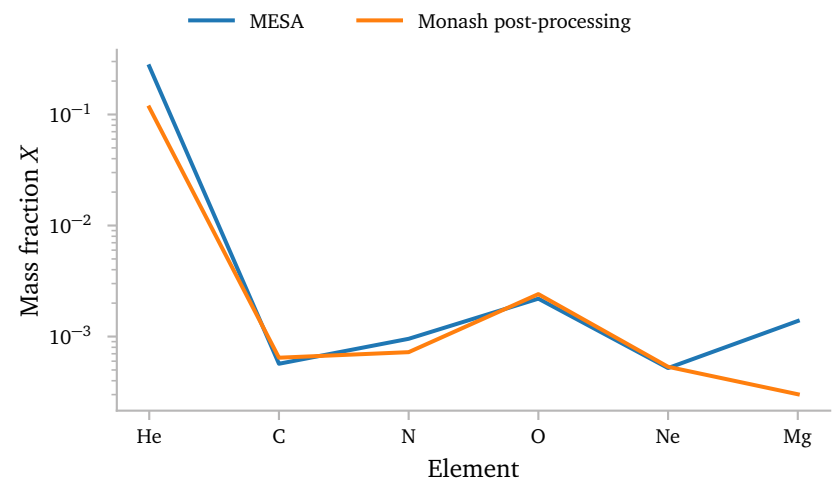


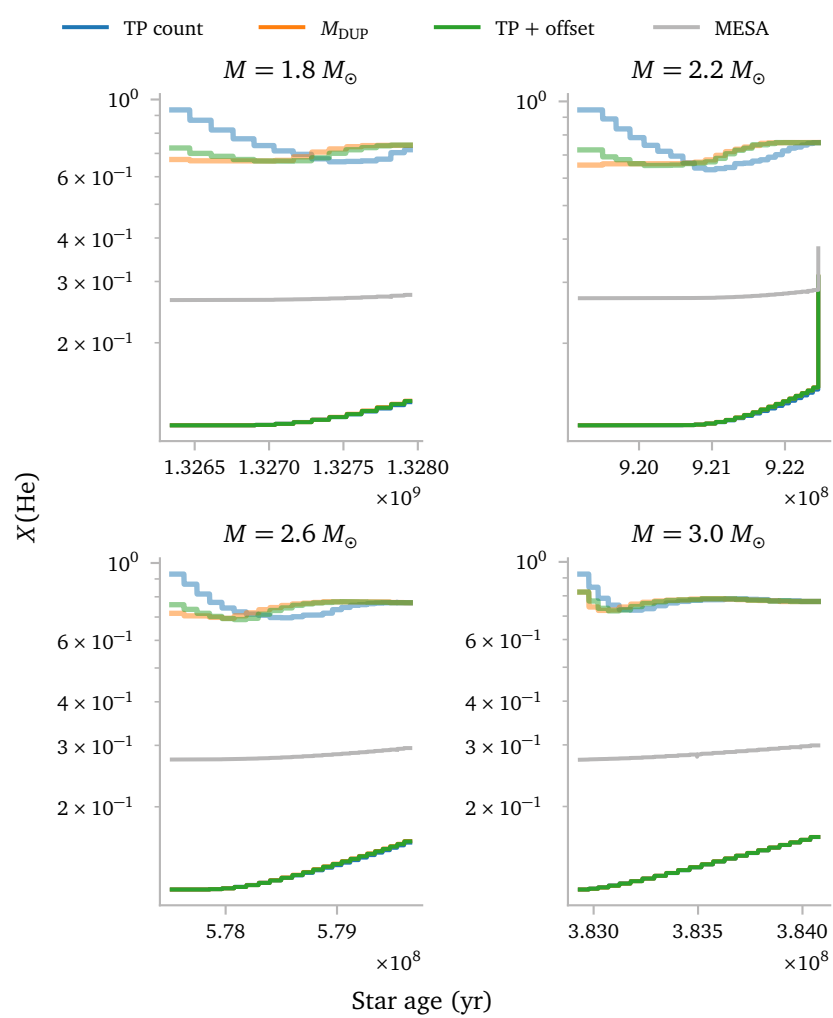
Notes Week 28

a. Monash Initial Helium Envelope Abundances

Last week, I showed these two plots:

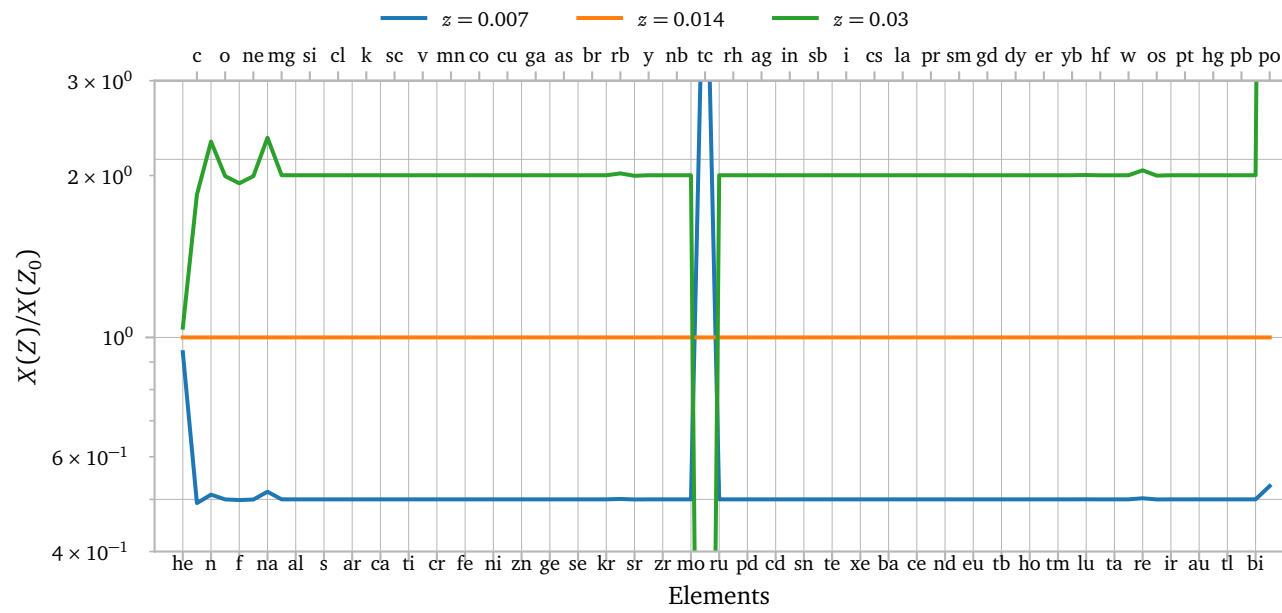


The most important thing we saw here was the quite large differences between the helium mass fractions.



Here, the same difference is again very clear.

It is actually very easy to understand where this discrepancy comes from, and in fact, I have already made a figure before that shows exactly what is going wrong here:

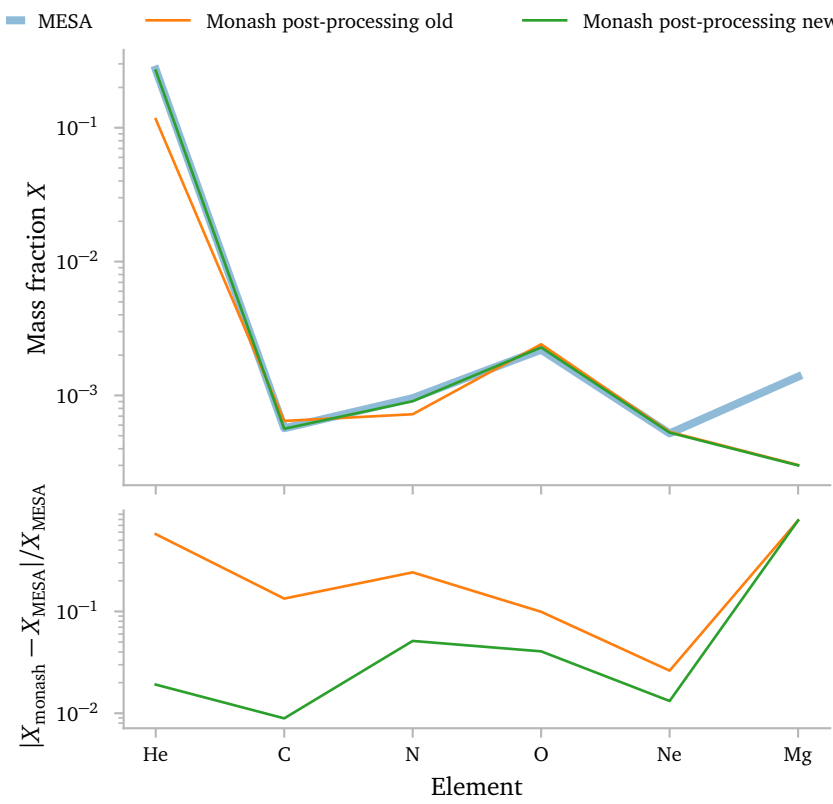


Here I plot the initial envelope abundances for 3 different Monash metallicities divided by the Monash solar Metallicity model.

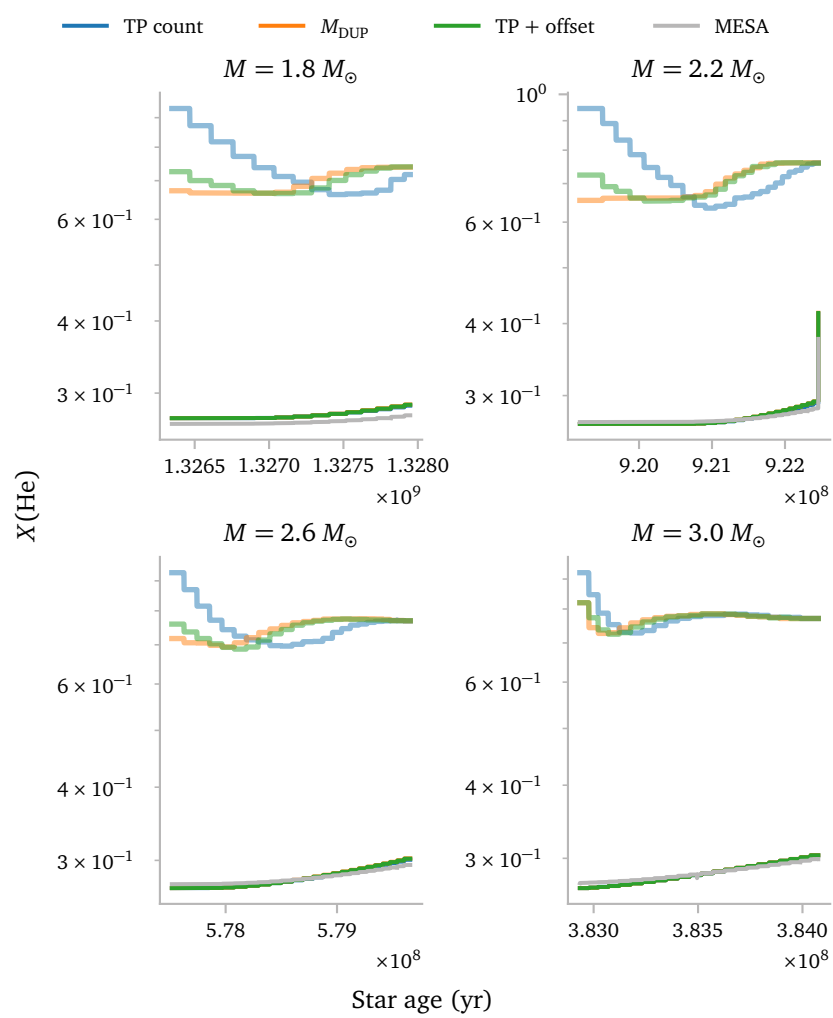
I made this plot to *confirm* I can simply take the abundance of an element, divide it by the Monash metallicity and multiply it by the MESA metallicity to get the MESA envelope abundance.

Now I think this works great for pretty much all elements, *except* for some of the very light elements, exactly the elements that I was comparing before.

I quickly fixed this by also interpolating the initial envelope abundances.



Clearly, this works much better for helium and actually all other isotopes too.



This looks much better!

b. MESA intershell abundances

In order to verify the Monash enrichment is working well, we can compare the enrichment of the envelope as predicted by my Monash post-processing code with the abundances from MESA.

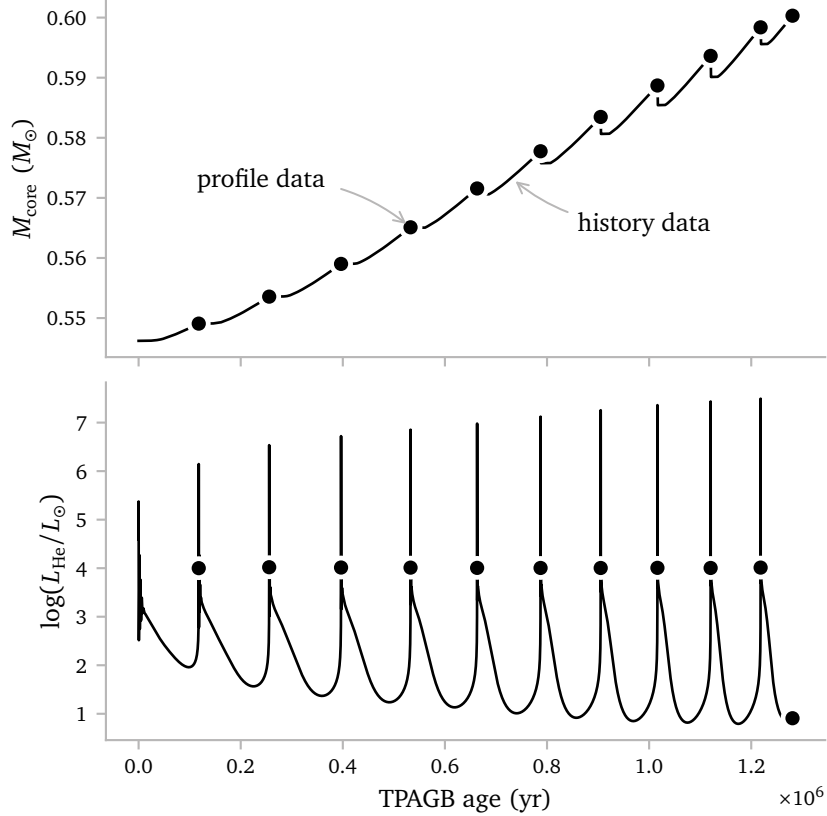
I think there are a couple interesting things to do. First, we just want to compare the intershell abundances of the elements that are available to me in the MESA models with the predicted Monash intershell abundances. This is already interesting, because this gives us a good idea of if the interpolation is working well.

Then, I think it is interesting to look at the envelope abundances to compare the differences. I have already done this previously, *but* now I have a “better” initial envelope abundance, and insight into the intershell abundances of both codes.

Lastly, I want to verify that my envelope enrichment code is working correctly. I will do this by injecting the MESA intershell abundances and the initial envelope abundance into my post-processor. I will then compare the envelope abundances it spits out with the actual MESA envelope abundances. If the post-processor is working well, it should reproduce the MESA envelope abundances well¹.

b.1 Intershell abundances

To start, I document *how* I obtain the intershell abundances using only a few profiles.



¹ Although I do not expect it to agree completely, because the dredge up assumed to be instantaneous, and the precise derived dredged up masses are slightly different from the MESA internal dredged-up masses.

For all single stars, I save a model and profile at the start of a thermal pulse, just before things really start to change and the helium burning is just beginning to start up.

These profiles contain a lot of relevant information, but they are sparse. To see *when* in the lifetime of stars the profiles are saved, I show the core mass and the helium luminosity.

Clearly, the models are correctly saved at the start of a thermal pulse. I think this is quite a nice time to sample the intershell abundance, since this is roughly also the abundance when dredge up occurs.

The last profile is an outlier. This is the model that is saved when the envelope mass is below 0.01 $M_{\odot}$.

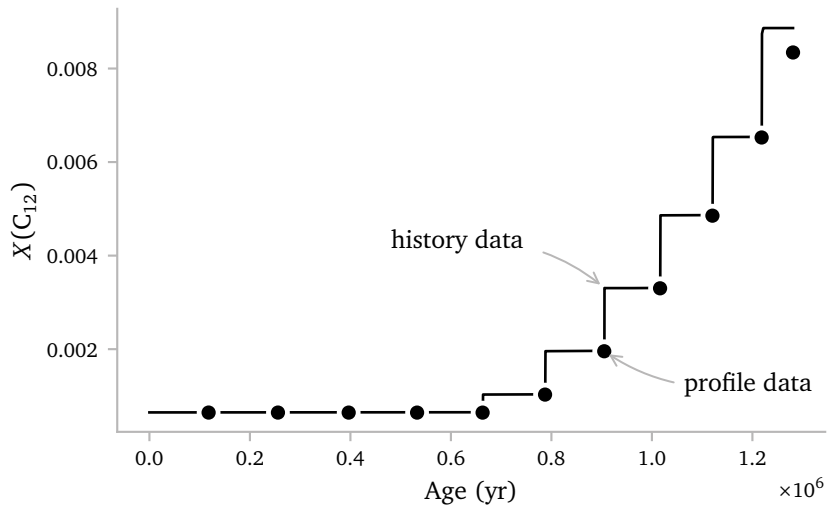
Now let me plot something a bit more interesting. By finding the position of the greatest change in the mass fraction of the metals²,

```
i = np.argmax(
    np.abs(
        np.log10(p.z_mass_fraction_metals[10:])
        - np.log10(p.z_mass_fraction_metals[:-10])
    )
)
```

we get a proxy for the intershell / envelope boundary.

By then simply taking the mass averaged abundance of an element a certain margin before this boundary, we get the envelope abundance. Since we also have my MESA implementation of the envelope abundances, we can compare the two.

² over a small number of zones



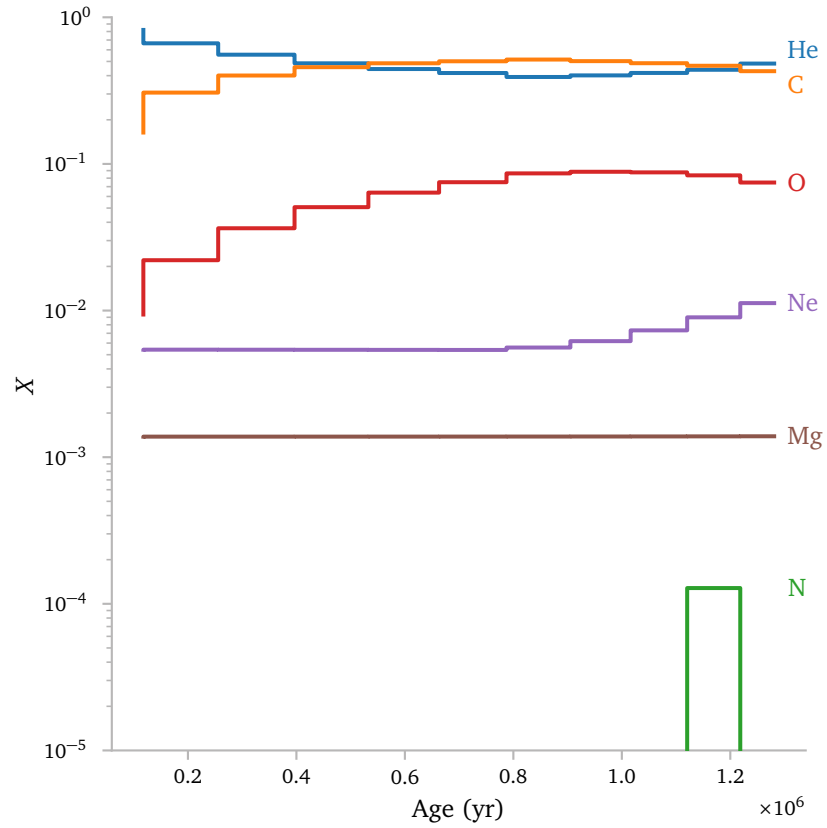
We clearly see that the two agree very well, even though they are computed with slightly different methods.

The MESA history data is based on the largest convective region. It then takes a mass average of this region. Luckily, and also unsurprisingly, this agrees well with the method that simply takes the mass averaged value of the abundances past the intershell zone.

The only difference occurs when the envelope mass is very small.

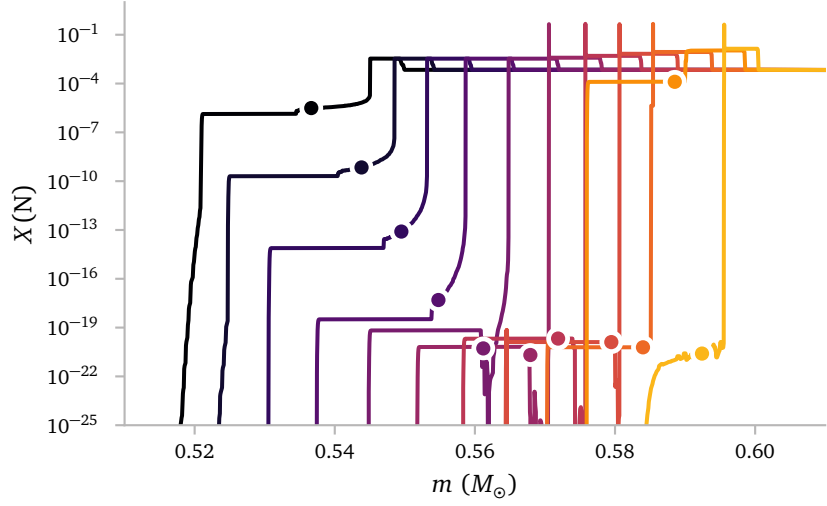
The shape of the envelope abundance, and very likely also the intershell abundance, is very nicely approximated by a step function.

Using the same criterion, but now taking some average to the intershell side of the intershell / envelope boundary, we can also find the intershell abundances. Unfortunately however, we cannot compare this with MESA history data.

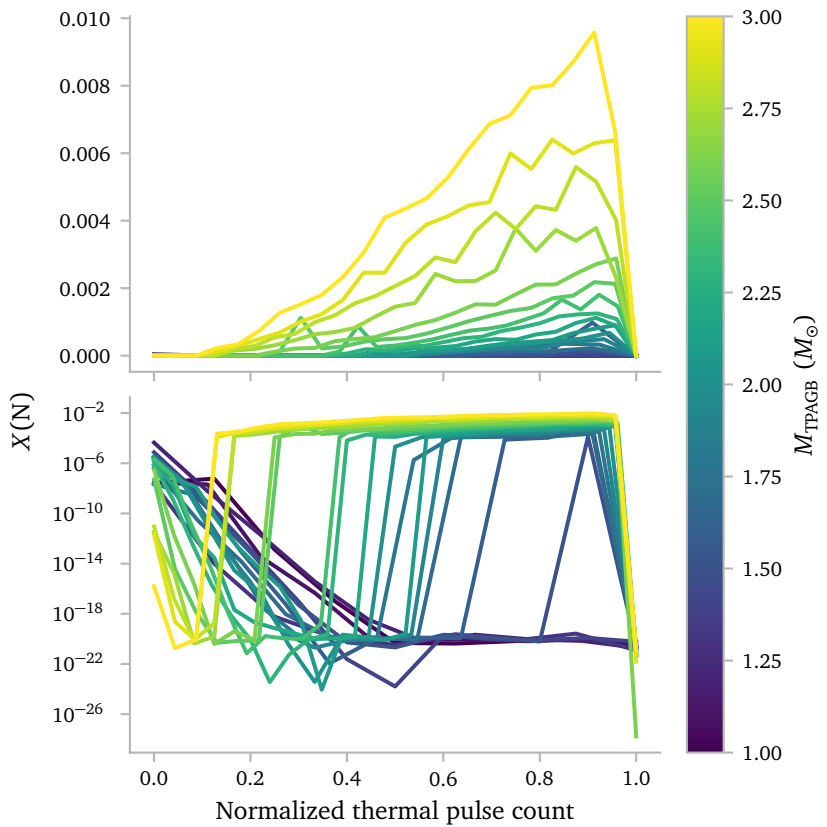


Here we can see all available elements together to see the relative abundances.

Everything looks normal to me, except for weird nitrogen bump. The nitrogen abundance is constantly



This is weird I think. But it is not necessarily something that is wrong with my intershell abundance finding code. There is just something that happens during this specific thermal pulse that makes the intershell relatively very abundant.



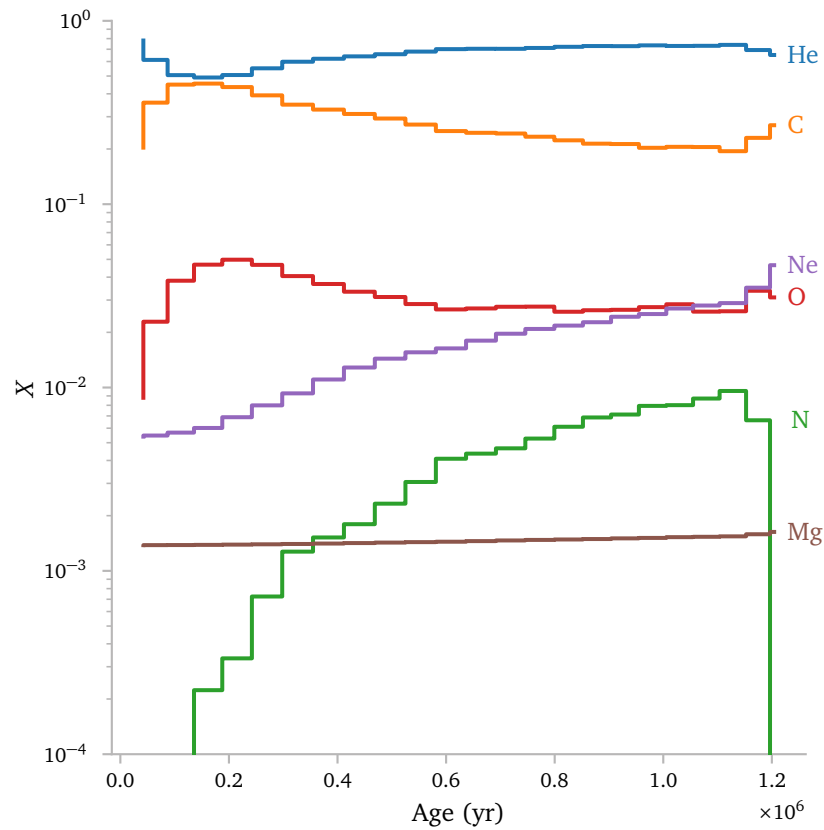
Apparently, the 1.5 $M_{\odot}$ model just by accident, only has 1 thermal pulse with a significant nitrogen abundance. This however does not mean that all stars have this 1 thermal pulse with a large nitrogen abundance, and it also does not seem like an error since there is quite a clear pattern.

Clearly, stars with masses larger than 1.5 $M_{\odot}$ have multiple pulses with a large nitrogen abundance. Stars with masses less than 1.5 $M_{\odot}$ have *no* thermal pulses with a large nitrogen abundance.

I do not know what causes these sudden increases in envelope abundances and why the abundances drop to low values again during the last thermal pulse.

Maybe it has something to do with the CNO cycle?

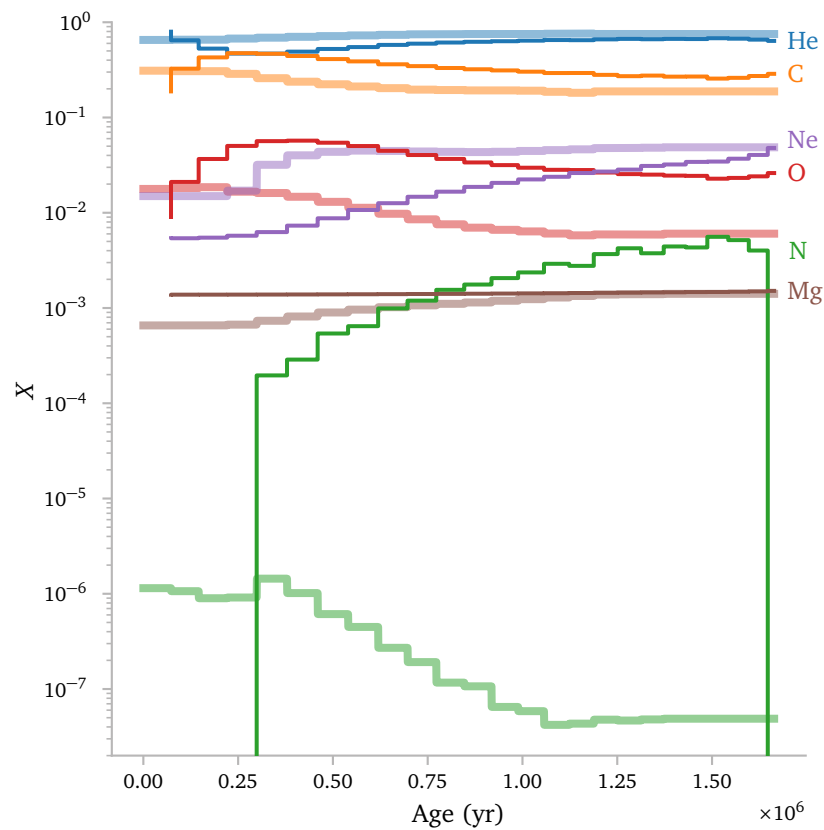
Obviously, the mass is quite important for the nitrogen abundance. Let me also plot the intershell abundances for the $3 M_{\odot}$ MESA model.



Obviously the nitrogen abundance is quite different for this $3 M_{\odot}$ model, but we see differences in the other isotopes too. Especially the neon abundance is much larger in this model.

Carbon seems *less* abundant in the heavier model.

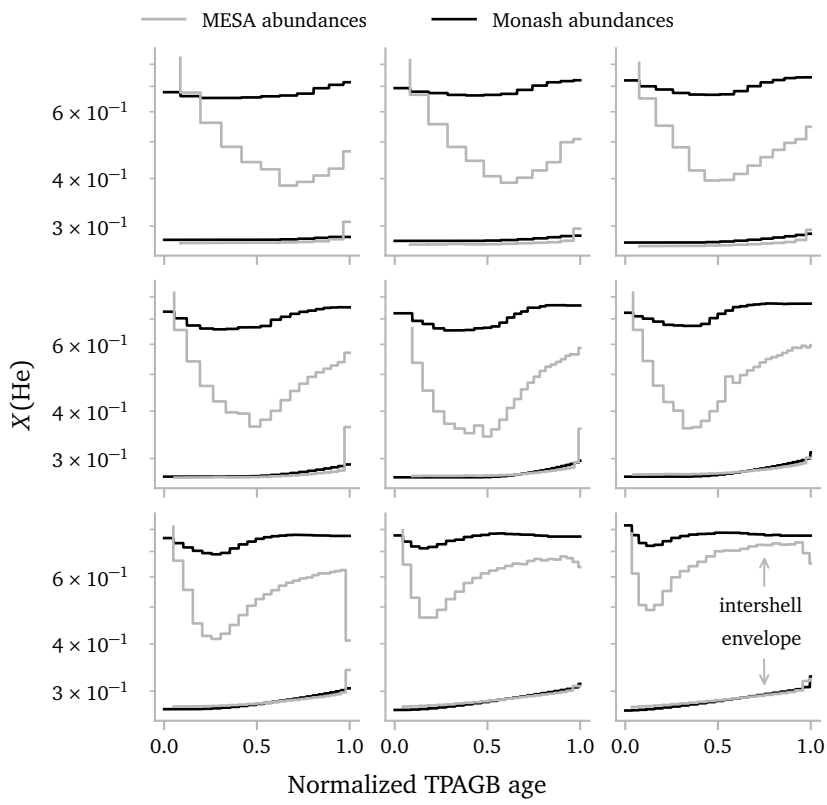
This figure below compares the MESA intershell abundances with the Monash post-processed intershell abundances for the $2.8 M_{\odot}$ model.



The nitrogen abundance is very different between the two codes. The magnesium abundances initially are a bit different, but end up at nearly the same value.

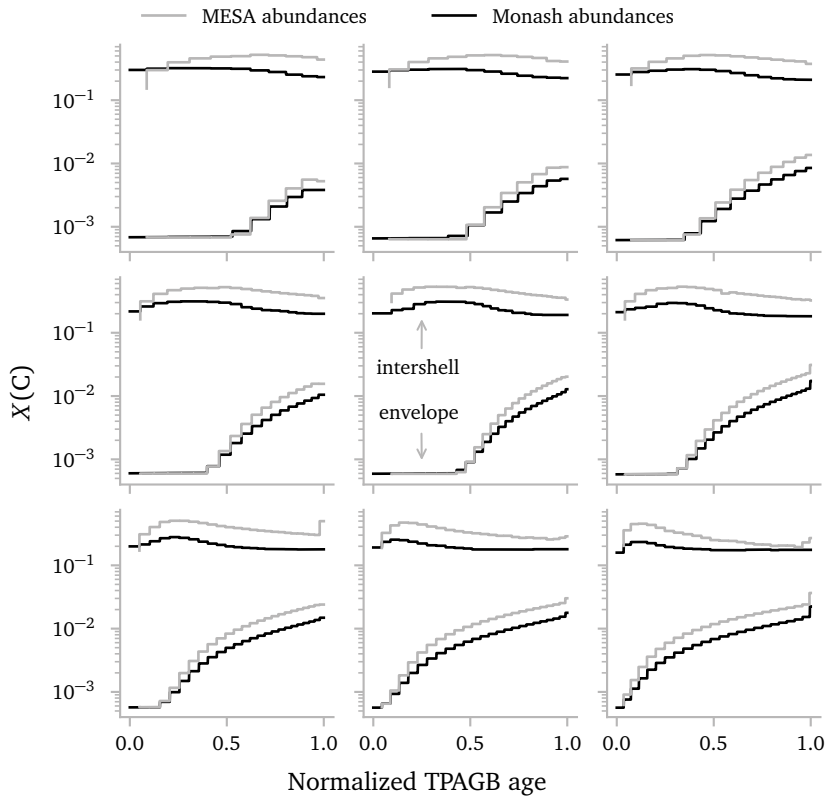
b.2 Envelope abundances

Now let me take a closer look at the intershell and envelope abundances for specific elements.



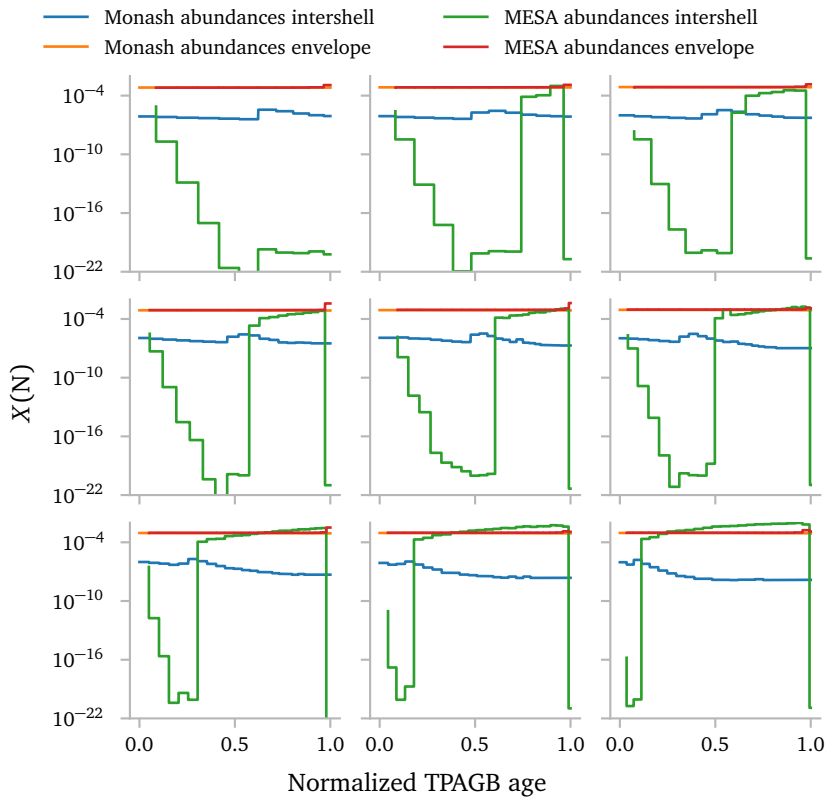
$M_{\text{TPAGB}} =$
1.4 1.6 1.8
2.0 2.2 2.4
2.6 2.8 3.0 $M_{\odot}$

The helium envelope abundances agree very well. Because of the new initial envelope abundance method, the initial values are very similar and because the intershell abundances are not much larger than the envelope abundances, dredge-up does not significantly impact the envelope abundances.



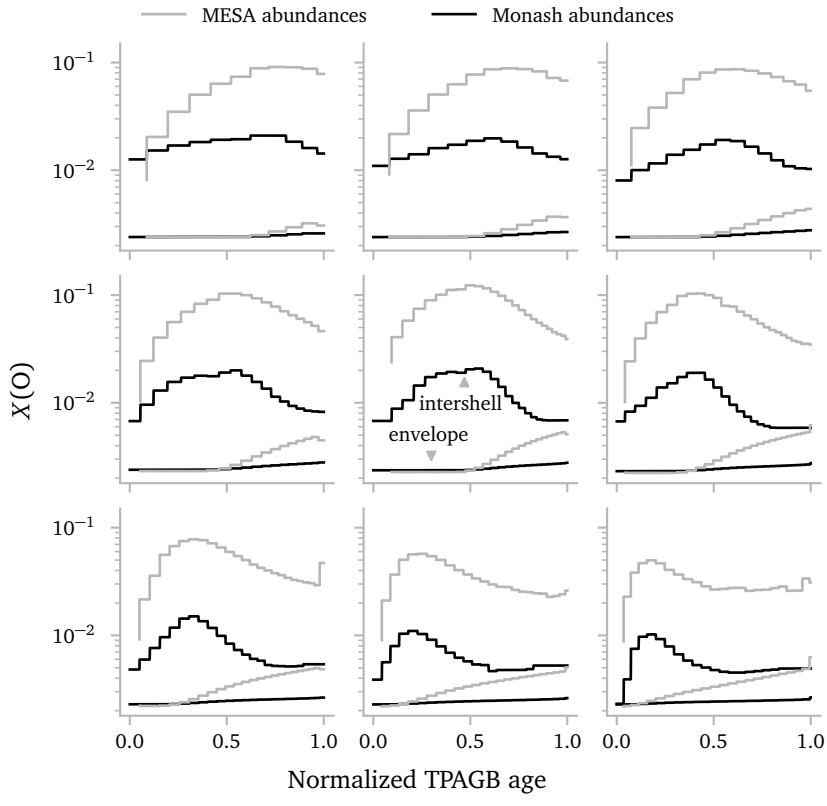
$M_{\text{TPAGB}} =$
1.4 1.6 1.8
2.0 2.2 2.4
2.6 2.8 3.0 $M_{\odot}$

The carbon abundances are a lot more interesting. The initial abundances are very similar. Just like last week however, we see that the carbon abundances in the envelope are lower for the Monash post-processed abundances. It makes sense because the intershell abundances are also lower for the Monash abundances.



$M_{\text{TPAGB}} =$
1.4 1.6 1.8
2.0 2.2 2.4
2.6 2.8 3.0 $M_{\odot}$

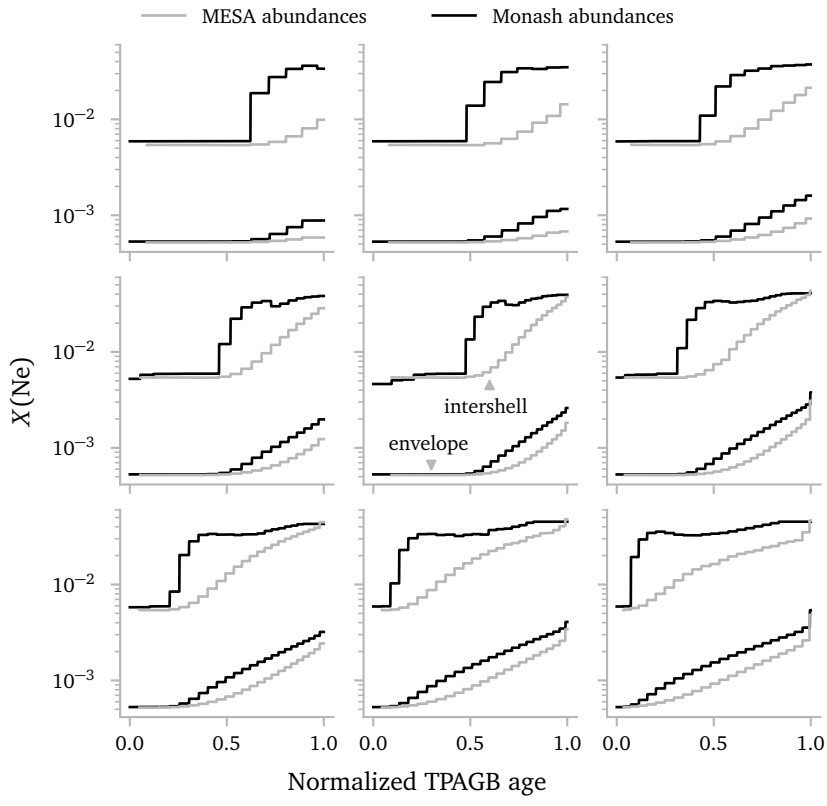
The intershell values are very different for the MESA and Monash abundances. I do not know why they are so different?



$M_{\text{TPAGB}} =$
1.4 1.6 1.8
2.0 2.2 2.4
2.6 2.8 3.0 $M_{\odot}$

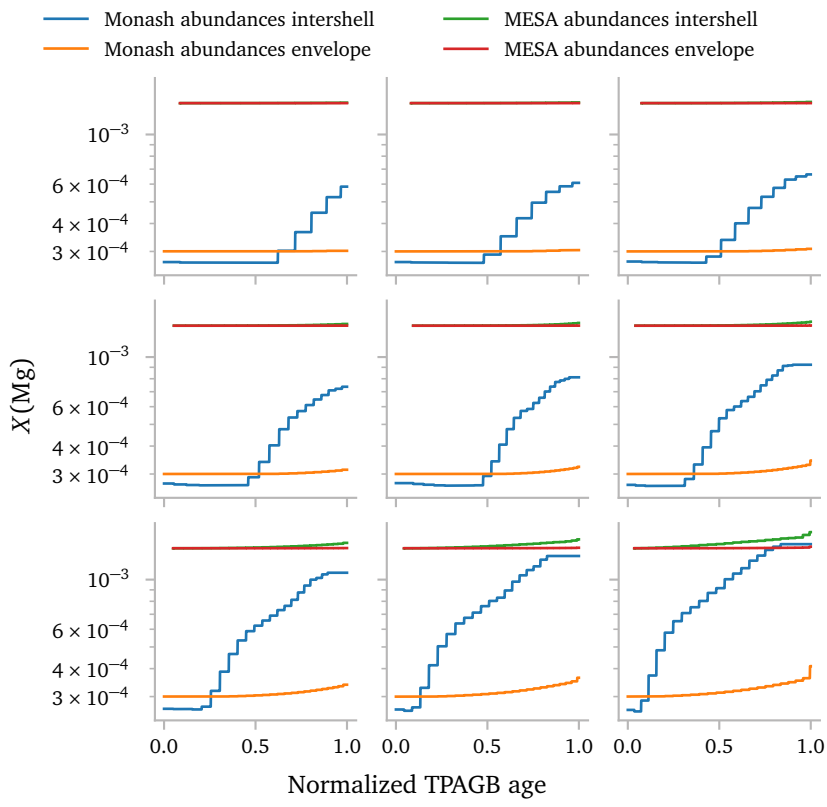
The initial oxygen envelope abundances once again agree very well. However, the intershell abundances are very different between the two codes. Because the Monash intershell abundances are much lower, the Monash post-processed envelope abundances do not increase with dredge-up as much as the MESA abundances.

The question is: why are the MESA intershell abundances so different from the Monash intershell abundances?



$M_{\text{TPAGB}} =$
1.4 1.6 1.8
2.0 2.2 2.4
2.6 2.8 3.0 $M_{\odot}$

I think the neon abundances are not very interesting. Overall both the envelope and intershell abundances agree pretty well. Just the shape of the Monash intershell abundance profiles rise earlier and more sharply which makes the Monash envelope abundances increase slightly more.

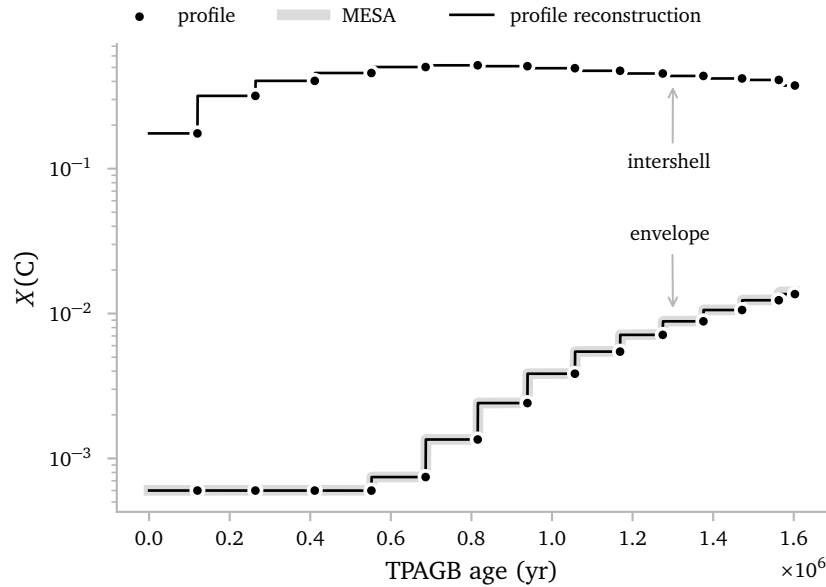


$M_{\text{TPAGB}} =$
1.4 1.6 1.8
2.0 2.2 2.4
2.6 2.8 3.0 $M_{\odot}$

The magnesium abundances are not very interesting because the initial MESA abundances are so much larger than the Monash initial abundances.

b.3 Post-processor with MESA abundances

Lastly, I want to test if my post-processing code reproduces the MESA envelope abundances if we supply it with the initial MESA envelope abundances with the MESA intershell abundances.

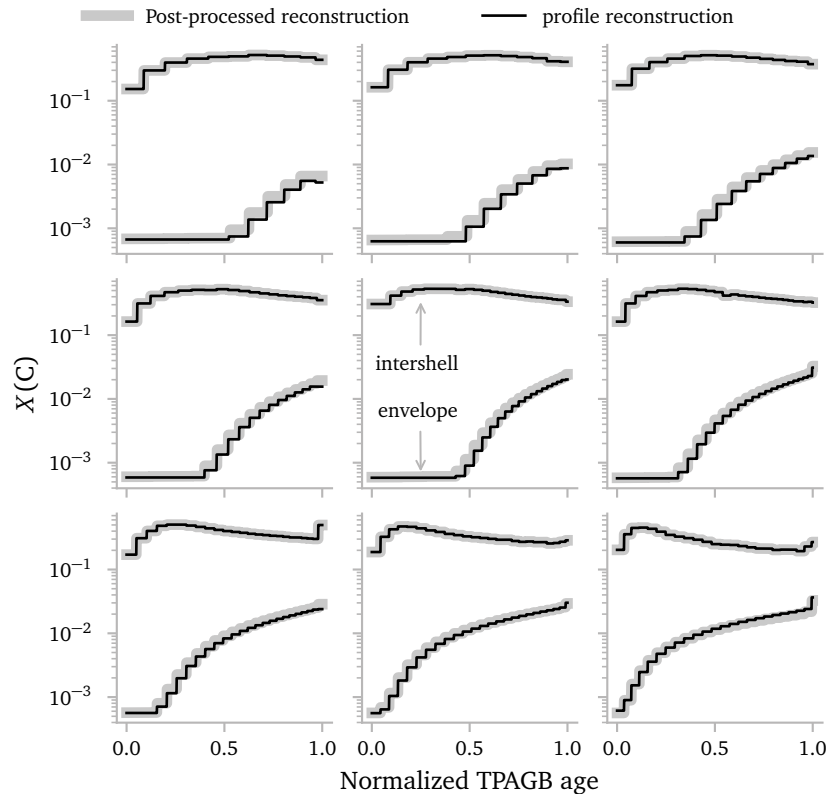


Firstly, I need to accurately reproduce the intershell abundance as a function of star model number. Now, since I only have the profiles, I need to make sure i correctly relate the sparse intershell abundances to the very densely spaces model numbers because my post-processing code expects this format.

I test that I correctly reproduce the envelope abundances.

Then I also produce the intershell values in the same way.

Now, let me give my post-processing code these intershell values and the initial envelope abundance.



$$M_{\text{TPAGB}} = \begin{matrix} 1.4 & 1.6 & 1.8 \\ 2.0 & 2.2 & 2.4 \\ 2.6 & 2.8 & 3.0 M_{\odot} \end{matrix}$$

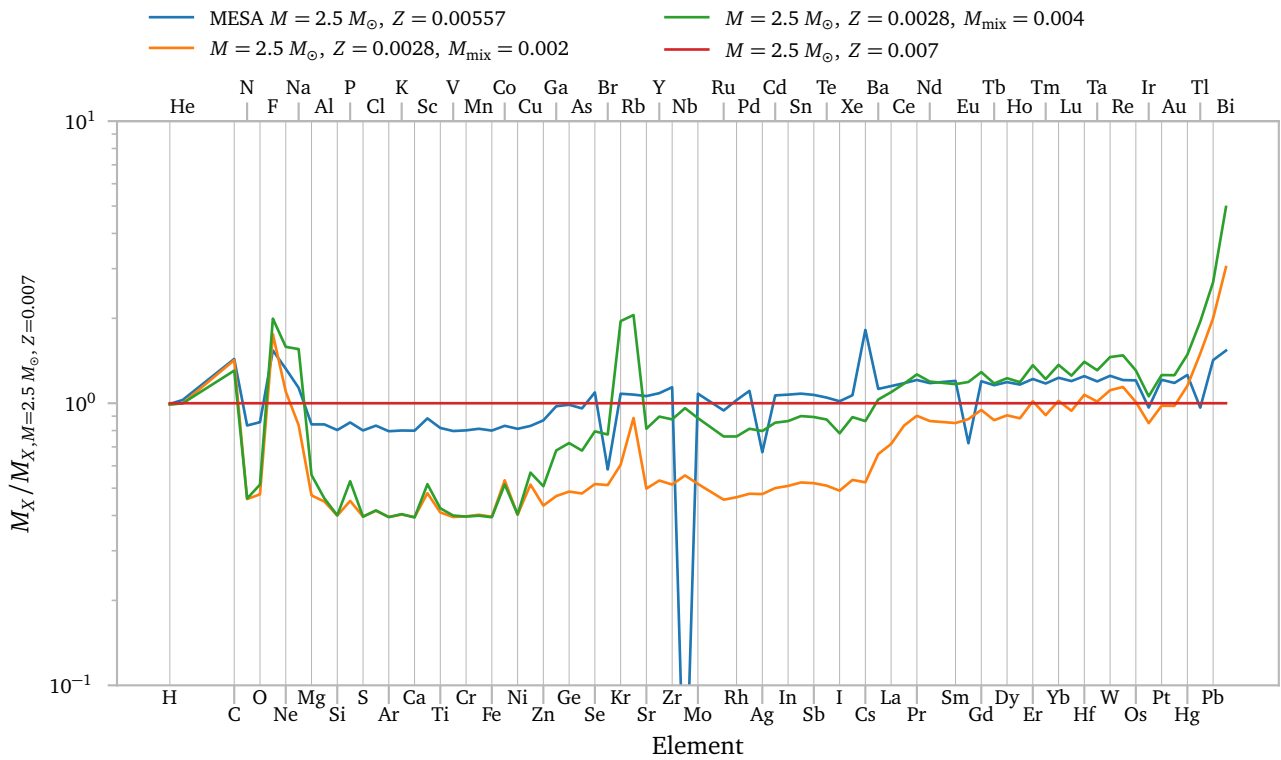
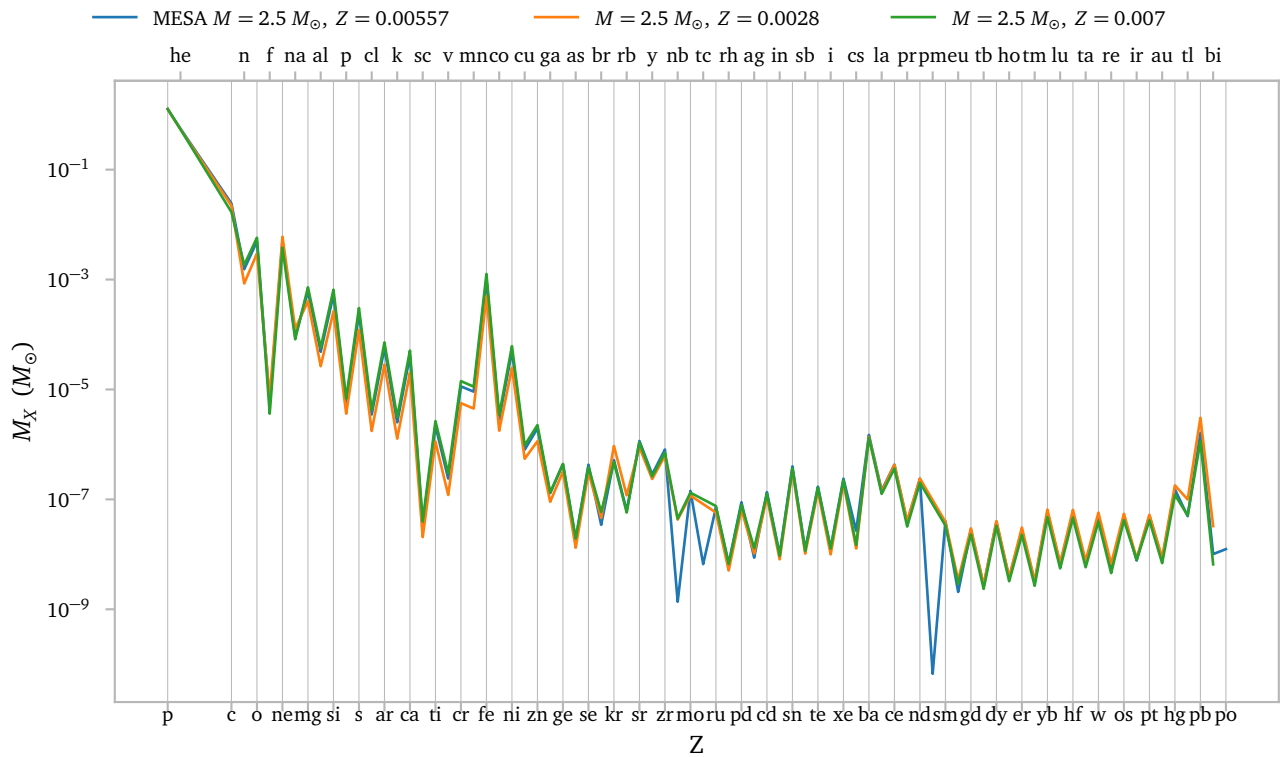
Clearly, the envelope value, *which* the post-processing code tries to reproduce, matches very well. This convinces me that my post-processor is correctly enriching the “fake” envelope.

This also means that the differences we saw in the plots above are only due to the differences in initial envelope abundance and intershell abundance evolution.

c. Investigating Monash Yields

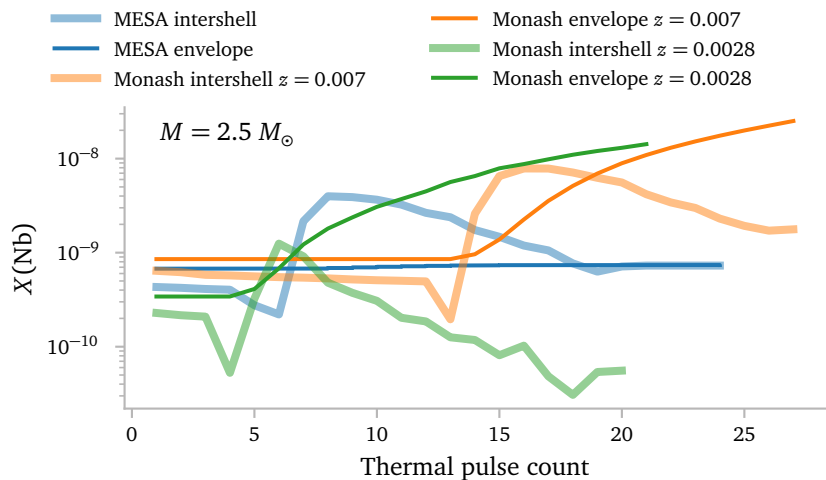
First things first, I made an error last week which quite drastically changed the results for the yields. However, the other results where all unaffected. The problem was in my implementation of the post-processor when it just uses the single star model. In this case, the post-processor does not have the parameters dict that the binary models do contain and the post-processor has to infer some of the relevant information itself.

Most importantly, it has to find the mass and the metallicity of the star. Now, I thought I would simply use the star property information for this, *but* apparently I had hardcoded a solar metallicity into the star model, which caused the post-processor to wrongly identify the models as solar metallicity models. It then used this metallicity to compute the interpolation and obtain the initial envelope abundances.



Compared to the plots from last week, this is *clearly* a much better match, and the offset is now actually.

There are however a few isotopes with big differences in the abundances between MESA and the different Monash models. One example is Niobium, where we see a very big drop in the yield for the MESA models.



d. Envelope abundances